

WHODUNNIT?

SKILL BUILDERS



**Up-and-moving
skill practice
activity!**

**Evaluating
Functions**



Whodunnit?

Evaluating Functions

Ideal Unit:	Relations, Functions & Patterns	Time Range:	20-40 Minutes	Supplies:	Pencil & Paper
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Common Core Alignment:

This particular activity was mapped to these common core standards. *Whodunnit* activities are ideal as a small group skill practice.

F-IF.2	Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.
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General Procedures:

A.) Give students a *Whodunnit Worksheet*. Students will need to solve the 10 problems about the topic. Each correct answer will result in a clue. Students will check off the Players, Last Known Whereabouts and Methods that are accounted for. The one remaining from each column is the correct solution. In this case it should be:

“Professor Delta used electrocution in the movie theater.”

B.) Print out the “Clues” and cut them out. I suggest you laminate them for future use. Spread them in different locations in the classroom and turn the students loose. Have teams of 2-3 circulate and attempt to figure out the clues. Once they think they have all the pieces, they can come to you to check their answer. If they are wrong, let them figure it out ☺

An alternative “Clue 10” is provided in the event you want a different result for different class periods.



Whodunnit?



Name : _____

Six contestants on a reality TV show were stunned to find their lowest scoring colleague was "injured." They must figure out the crime before the bell rings. The question is *Whodunnit?* And how... The Player, Last Known Whereabouts and Method that are left unaccounted for -- is the solution.

1.	2.
3.	4.
5.	6.

7.

8.

9.

10.

Who is the "Criminal"?

(The three boxes left unchecked will reveal the crime. If you cannot figure it out, you may be next.)

The Players

Dr. Alpha	☐
Miss Beta	☐
Mrs. Gamma	☐
Professor Delta	☐
Mr. Epsilon	☐
Coach Omega	☐

**The Last Known
Whereabouts**

Atrium	☐
Gym	☐
Kitchen	☐
Library	☐
Movie Theater	☐
Pool	☐

The Method

Chemical Poisoning	☐
Cougar Attack	☐
Electrocution	☐
Fallen Object	☐
Stubbed Toe	☐
Venomous Bite	☐

???

Clue #1

W E O D U N N I I ?

If $f(x) = 2x + 5$, what is $f(3)$?

16 Dr. Alpha was working on free throws in the gym.

6 The victim was poisoned with any chemicals.

10 The victim did not stub a toe.

11 Mr. Epsilon was mopping the kitchen floor.

W E O D U N N I I ?

Clue #2

If $g(x) = \frac{3x}{2}$, what is $g(-4)$?

6 Professor Delta was swimming laps in the pool.

-6 A fallen object was not the cause of the injury.

-8 The injury was not from an attack by a cougar.

-12 Mrs. Gamma was watching *Despicable You* in the movie theater.

???

???

Clue #3

WFOODUNNIT?

If $h(n) = \frac{11n - 14}{3n - 2}$, what is $h(6)$?

$\frac{8}{3}$

A venomous snake did not bite the victim.

-3

Coach Omega was doing pushups in the atrium.

$\frac{59}{3}$

Electrocution did not injure the victim.

$\frac{13}{4}$

Dr. Alpha was reading periodicals in the library.

WFOODUNNIT?

Clue #4

If $y(x) = |x + 3| - 4$, what is $y(-10)$?

3

Chemical poisoning was not the cause of the injury.

-11

A cougar did not injure the victim.

9

Coach Omega was admiring clouds while sitting by the pool.

11

Miss Beta was doing Tai Chi in the gym.

???

???

Clue #5

WFOODUNNIT?

If $s(t) = 4^{3t}$, what is $s(-1)$?

64 Professor Delta was reading *10 Ways to Think Smarter* in the library.

$\frac{1}{64}$ Miss Beta was pressure washing the pool cover.

16 A stubbed toe was not the injury.

27 The victim was crushed by a fallen object.

WFOODUNNIT?

Clue #6

If $g(x) = 4x + 14$, what is $g(x - 3)$?

$4x + 2$ The victim did not stub a toe.

$x - 3$ Professor Delta was performing physics experiments in the gym.

$4x - 11$ Mrs. Gamma was making popcorn in the movie theater.

$4x + 8$ Electrocution was not the cause of the injury.

???

???

Clue #7

W E O D U N N I I ?

If $h(t) = -16t^2 + 40$, what is $h(3t)$?

$-144t^2 + 40$ The victim did not get injured from a venomous bite.

$-48t^2 + 120t$ Mr. Epsilon was checking his email in the library.

$-144t + 40$ Mrs. Gamma was reading in the kitchen

$-48t^2 + 40$ The victim did not stub a toe.

W E O D U N N I I ?

Clue #8

If $f(x) = x^2 - 4$, what is $f(x + 3)$?

$x^2 - 1$ The victim was not crushed by a fallen object.

$x^2 + 6x + 5$ The victim was not mauled by a cougar.

$x^2 + x - 1$ Miss Beta was organizing bookshelves in the library.

$x^2 + 5$ Mr. Epsilon was inspecting the equipment in the gym.

???

???

Clue #9

W E O D U N N I I ?

If $v(t) = \frac{t^2 - t}{3t + 1}$, what is $v(t^3)$?

$$t^6 - t^3$$

A stubbed toe was not the injury.

$$\frac{t^3 - 1}{4}$$

The victim was not electrocuted.

$$\frac{t^5 - t^3}{3t^3 + 1}$$

Professor Delta was installing a new projector in the movie theater.

$$\frac{t^6 - t^3}{3t^3 + 1}$$

Mrs. Gamma was doing Yoga in the gym.

W E O D U N N I I ?

Clue #10

If $d(3) = 4$ and $d(x) = 7x - k$, what is k ?

7

The victim did not stub a toe.

25

Professor Delta was performing physics experiments in the movie theater.

17

Coach Omega was eating popcorn in the atrium.

3

Electrocution was not the cause of the injury.

???

???

Alternate Ending

W E O D U N N I I ?

"Coach Omega used electrocution in the atrium".

W E O D U N N I I ?

Clue #10

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- | | |
|----|--|
| 7 | The victim did not stub a toe. |
| 25 | Coach Omega was eating popcorn in the atrium. |
| 17 | Professor Delta was performing physics experiments in the movie theater. |
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Alternate Ending

W E O D U N N U ?

"Coach Omega used electrocution in the atrium".

W E O D U N N U ?

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